

Johansen Cointegration Test

Date: 07/09/26 Time: 12:00
Sample: 1/04/2010 12/30/2025
Included observations: 4159
Lags interval (in first differences): 1 to 4
Endogenous variables: JPYUSD CNYUSD
Deterministic assumptions: Case 3 (Johansen-Hendry-Juselius): Cointegrating relationship includes a constant. Short-run dynamics include a constant.

Unrestricted Cointegration Rank Test (Trace)

Hypothesized No. of CE(s)	Eigenvalue	Trace Statistic	0.05 Critical Value	Prob.** Critical Value
None *	0.002995	15.51923	15.49471	0.0496
At most 1	0.000736	3.057494	3.841465	0.0804

Trace test indicates 1 cointegrating equation(s) at the 0.05 level

* denotes rejection of the hypothesis at the 0.05 level

**MacKinnon-Haug-Michelis (1999) p-values

Unrestricted Cointegration Rank Test (Max-eigenvalue)

Hypothesized No. of CE(s)	Eigenvalue	Max-Eigen Statistic	0.05 Critical Value	Prob.** Critical Value
None	0.002995	12.46173	14.26460	0.0945
At most 1	0.000736	3.057494	3.841465	0.0804

Max-eigenvalue test indicates no cointegration at the 0.05 level

* denotes rejection of the hypothesis at the 0.05 level

**MacKinnon-Haug-Michelis (1999) p-values

Unrestricted Cointegrating Coefficients (norma

JPYUSD	CNYUSD
587.7911	-132.8244
333.9726	66.39806

Unrestricted Adjustment Coefficients (alpha):

D(JPYUSD)	1.51E-06	-1.26E-06
D(CNYUSD)	2.41E-05	5.82E-06

1 Cointegrating Equation
Log-Likelihood: 60717.37

Normalized cointegrating coeffici

JPYUSD	CNYUSD
1.000000	-0.225972
	(0.05947)

Adjustment coefficients (standard

D(JPYUSD)	0.000887
	(0.00049)
D(CNYUSD)	0.014150
	(0.00446)